

Class 11: Alphafold

Mayla (19136116)

```
library(bio3d)

pdbfiles <- list.files("hivpr_dimer_94b5b.result",
                      pattern = "\\..pdb$",
                      recursive = TRUE,
                      full.names = TRUE)

pdbfiles
```

```
[1] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_1"
[2] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_2"
[3] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3"
[4] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_4"
[5] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_5"
```

```
pdb <- read.pdb(pdbfiles[1])
pdb
```

Call: read.pdb(file = pdbfiles[1])

Total Models#: 1

Total Atoms#: 757, XYZs#: 2271 Chains#: 1 (values: A)

Protein Atoms#: 757 (residues/Calpha atoms#: 99)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 0 (residues: 0)

Non-protein/nucleic resid values: [none]

Protein sequence:

PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD

QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF

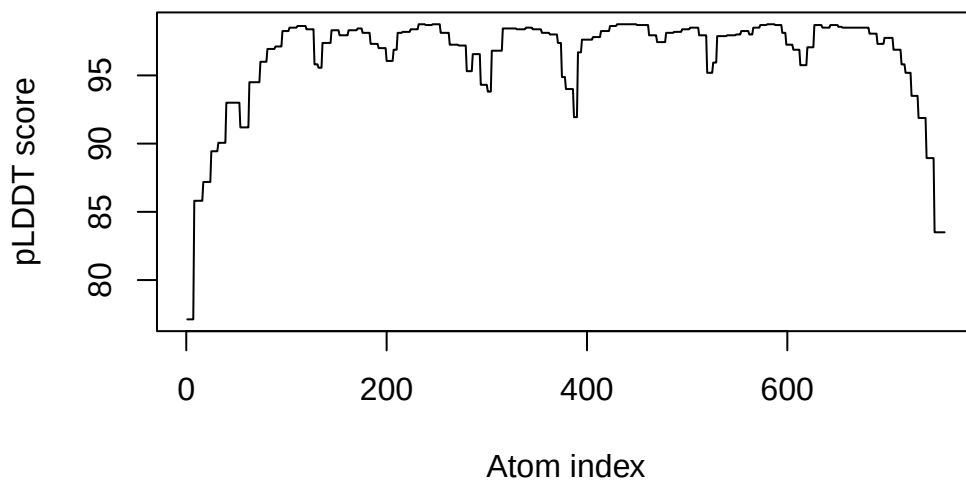
+ attr: atom, xyz, calpha, call

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	-15.227	-6.625	-5.617	1	77.12
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	-15.734	-6.305	-6.949	1	77.12
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	-14.945	-6.984	-8.062	1	77.12
4	ATOM	4	CB	<NA>	PRO	A	1	<NA>	-15.594	-4.781	-7.027	1	77.12
5	ATOM	5	O	<NA>	PRO	A	1	<NA>	-13.750	-7.246	-7.906	1	77.12
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	-14.492	-4.453	-6.066	1	77.12

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	C	<NA>
5	<NA>	O	<NA>
6	<NA>	C	<NA>

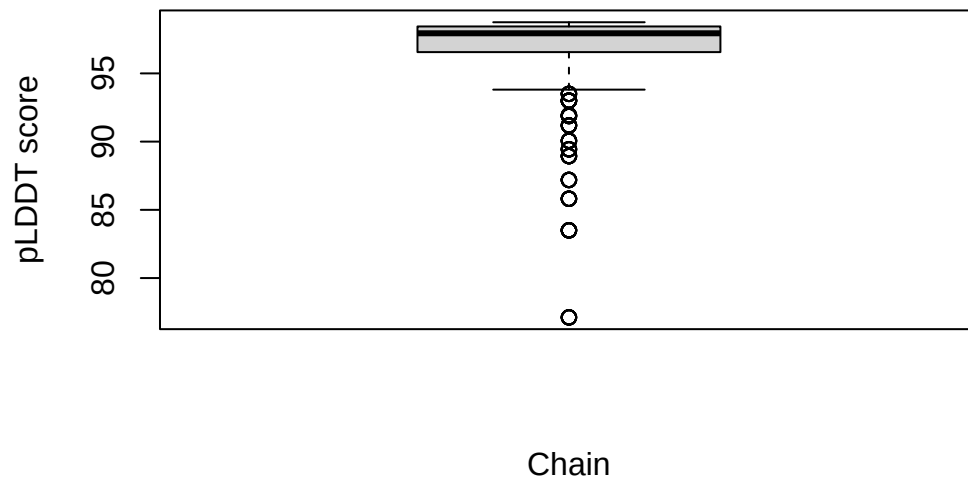
```
plot(pdb$atom$b,  
      type = "l",  
      xlab = "Atom index",  
      ylab = "pLDDT score")
```



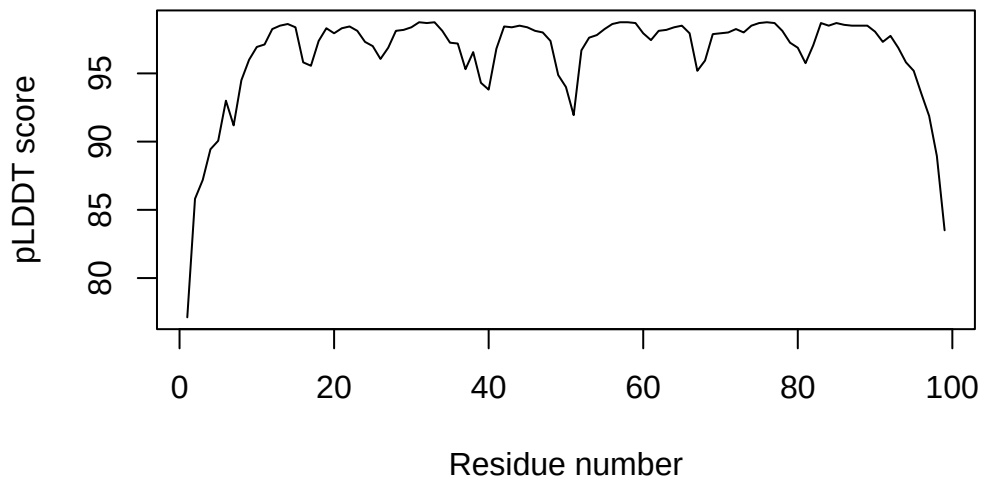
```
table(pdb$atom$chain)
```

```
A  
757
```

```
boxplot(pdb$atom$b ~ pdb$atom$chain,  
        xlab = "Chain",  
        ylab = "pLDDT score")
```



```
ca.inds <- atom.select(pdb, elety = "CA")  
  
ca.data <- pdb$atom[ca.inds$atom, ]  
  
plot(ca.data$resno,  
      ca.data$b,  
      type = "l",  
      xlab = "Residue number",  
      ylab = "pLDDT score")
```



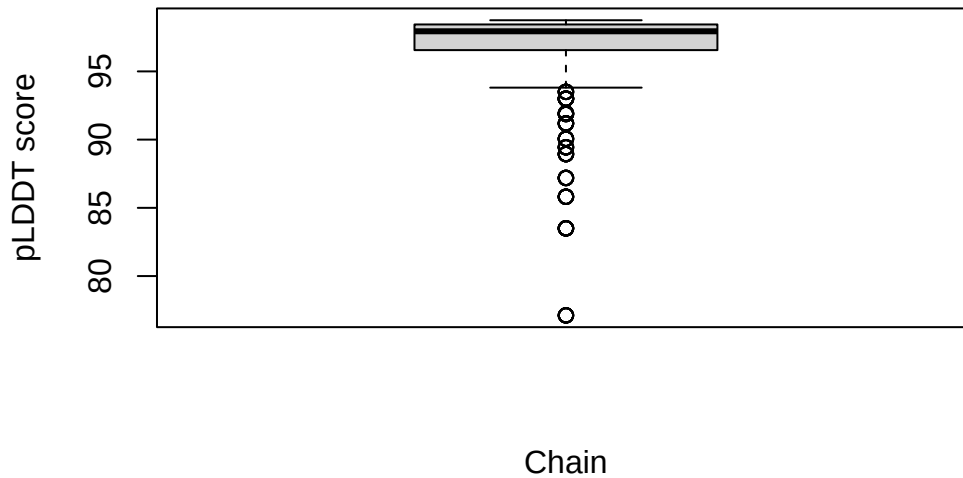
The AlphaFold model has generally high confidence where pLDDT values are high. Lower pLDDT regions are less reliable and may correspond to flexible or disordered parts of the structure. The two chains can be compared using the boxplot to see whether one chain has lower confidence than the other.

```
table(pdb$atom$chain)
```

```
A  
757
```

```
boxplot(pdb$atom$b ~ pdb$atom$chain,  
        xlab = "Chain",  
        ylab = "pLDDT score",  
        main = "pLDDT Scores by Chain")
```

pLDDT Scores by Chain



The pLDDT scores compare AlphaFold confidence between the two chains. Higher pLDDT values indicate more reliable predicted regions, while lower values suggest less confident or more flexible regions.

```
#library(NGLViewer)

#NGLViewer(pdbfiles[1]) |>
#  addRepresentation("cartoon")

#Change this for YOUR results dir name
results_dir <- "hivpr_dimer_94b5b.result/"

files <- list.files(results_dir)
files

[1] "test_94b5b"

scores_files <- list.files(
  path = paste0(results_dir, files[1]),
  pattern = "scores",
  full.names = TRUE
)
scores_files
```

```
[1] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_001_alphafold2_ptm_model_4_s
[2] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_002_alphafold2_ptm_model_5_s
[3] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_003_alphafold2_ptm_model_3_s
[4] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_004_alphafold2_ptm_model_1_s
[5] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_005_alphafold2_ptm_model_2_s
```

```
rid = "k1w3m8"
```

```
library(jsonlite)
scores <- lapply(scores_files, read_json)
```

```
rid="j8n2vp"
```

```
names(scores)
```

```
NULL
```

```
rid="d9q7m"
```

```
pae <- scores[[1]]$pae
```

```
library(bio3d)
```

```
results_dir <- "hivpr_dimer_94b5b.result/"
```

```
pdb.files <- list.files(path = results_dir,
                        pattern = ".pdb",
                        recursive = TRUE,
                        full.names = TRUE)
```

```
basename(pdb.files)
```

```
[1] "test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb"
[2] "test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb"
[3] "test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb"
[5] "test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb"
```

```
pdbbs <- pdbaln(pdb.files, fit = TRUE, exefile = "msa")
```

Reading PDB files:

```
hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_see
hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_see
hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_see
hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_see
hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_see
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_001_alphafold2_ptm_model_4_see
pdb/seq: 2   name: hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_002_alphafold2_ptm_model_5_see
pdb/seq: 3   name: hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_003_alphafold2_ptm_model_3_see
pdb/seq: 4   name: hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_004_alphafold2_ptm_model_1_see
pdb/seq: 5   name: hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_unrelaxed_rank_005_alphafold2_ptm_model_2_see
```

pdbs

```

1                               .                               .                               .                               .                               50
[Truncated_Name:1]test_94b5b  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]test_94b5b  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]test_94b5b  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]test_94b5b  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]test_94b5b  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
*****
1                               .                               .                               .                               .                               50

51                               .                               .                               .                               .                               99
[Truncated_Name:1]test_94b5b  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]test_94b5b  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]test_94b5b  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]test_94b5b  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]test_94b5b  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
51                               .                               .                               .                               .                               99
```

Call:

```
pdbaln(files = pdb.files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```


Alignment dimensions:

5 sequence rows; 99 position columns (99 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call

```
rd <- rmsd(pdbbs, fit = TRUE)
```

Warning in rmsd(pdbbs, fit = TRUE): No indices provided, using the 99 non NA positions

```
range(rd)
```

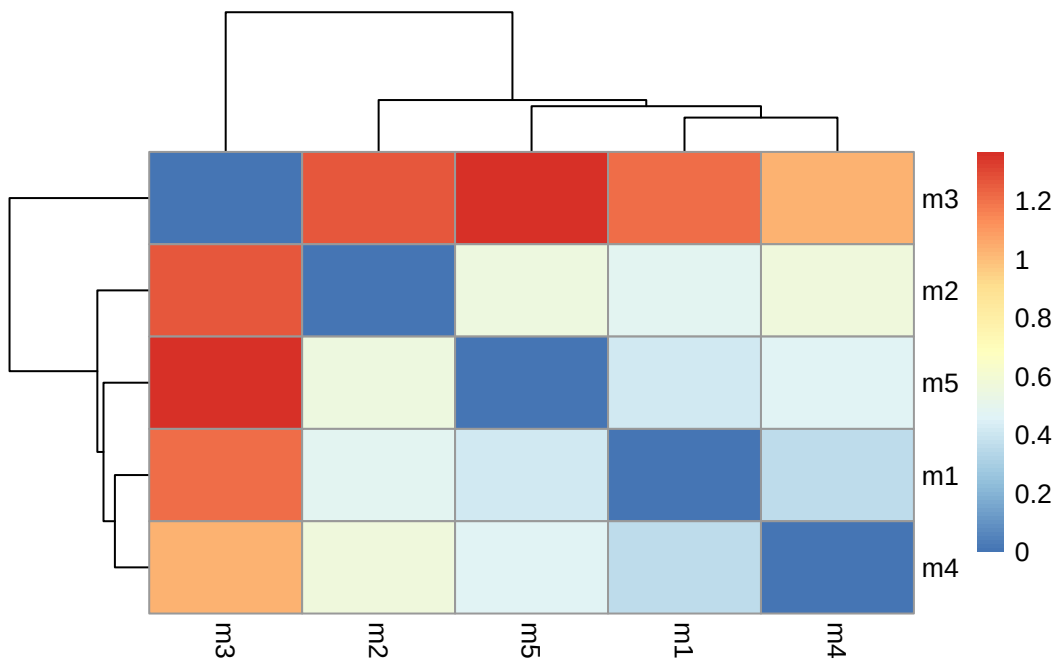
```
[1] 0.000 1.367
```

```
library(pheatmap)
```

```
colnames(rd) <- paste0("m", 1:5)
```

```
rownames(rd) <- paste0("m", 1:5)
```

```
pheatmap(rd)
```



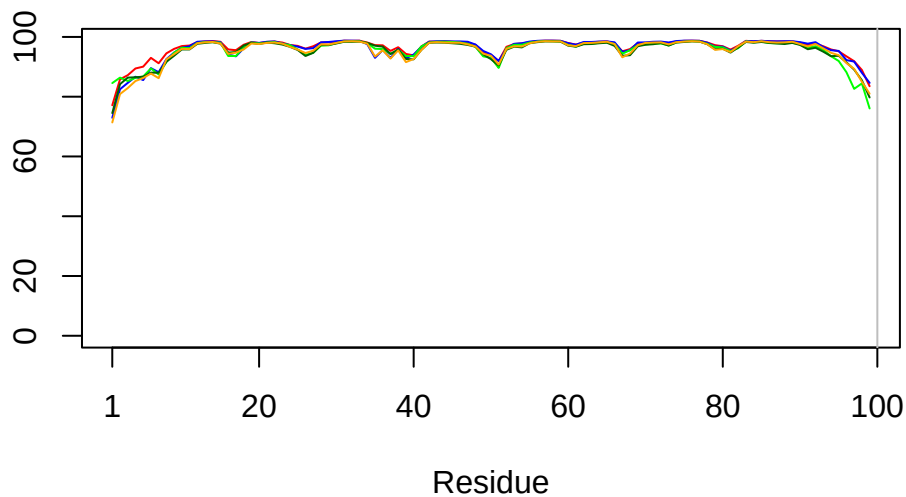
```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ = "l", col = "red", sse = pdb)
```

Warning in plotb3(pdb\$b[1,], typ = "l", col = "red", sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
points(pdb$b[2,], typ = "l", col = "blue")
points(pdb$b[3,], typ = "l", col = "green")
points(pdb$b[4,], typ = "l", col = "darkgreen")
points(pdb$b[5,], typ = "l", col = "orange")
abline(v = 100, col = "gray")
```



```
core <- core.find(pdb)
```

```
core size 98 of 99  vol = 4.608
core size 97 of 99  vol = 3.729
core size 96 of 99  vol = 2.981
```

```
core size 95 of 99  vol = 2.312
core size 94 of 99  vol = 1.831
core size 93 of 99  vol = 1.359
core size 92 of 99  vol = 0.992
core size 91 of 99  vol = 0.606
core size 90 of 99  vol = 0.38
FINISHED: Min vol ( 0.5 ) reached
```

```
core.inds <- print(core, vol = 0.5)
```

```
# 91 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     3   3      1
2     7  96     90
```

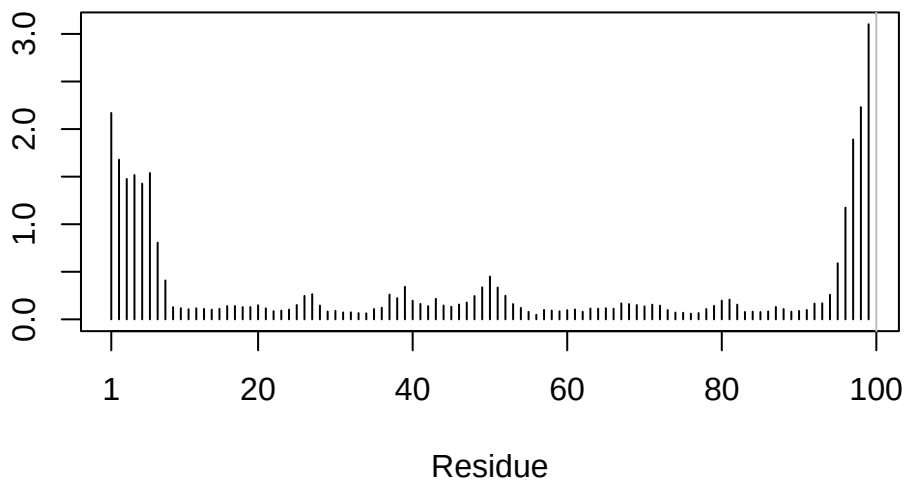
```
xyz <- pdbfit(pdb, core.inds, outpath = "corefit_structures")
```

```
rf <- rmsf(xyz)
```

```
plotb3(rf, sse = pdb)
```

Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
abline(v = 100, col = "gray", ylab = "RMSF")
```



```
library(jsonlite)
```

```
pae_files <- list.files(path = results_dir,
                        pattern = "model.*\\.json",
                        recursive = TRUE,
                        full.names = TRUE)
```

```
pae_files
```

```
[1] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_001_alphafold2_ptm_model_4_s
[2] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_002_alphafold2_ptm_model_5_s
[3] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_003_alphafold2_ptm_model_3_s
[4] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_004_alphafold2_ptm_model_1_s
[5] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_scores_rank_005_alphafold2_ptm_model_2_s
```

```
pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
```

```
pae5 <- read_json(pae_files[5], simplifyVector = TRUE)
```

```
attributes(pae1)
```

```
$names
```

```
[1] "plddt" "max_pae" "pae" "ptm"
```

```
head(pae1$plddt)
```

```
[1] 77.12 85.81 87.19 89.44 90.06 93.00
```

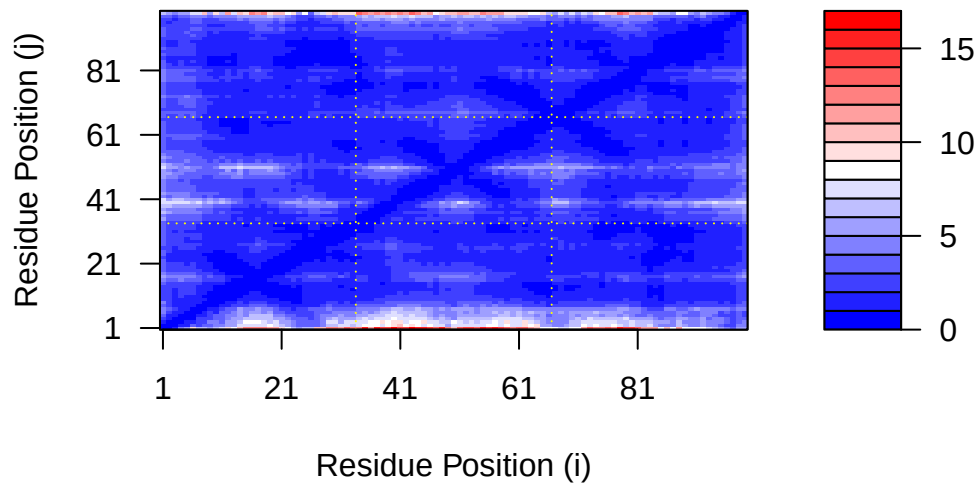
```
pae1$max_pae
```

```
[1] 16.96875
```

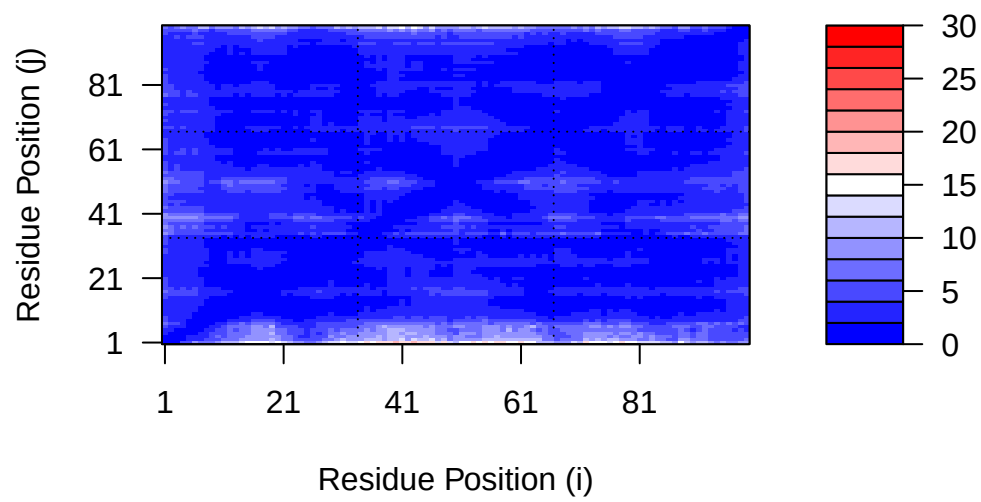
```
pae5$max_pae
```

```
[1] 19.3125
```

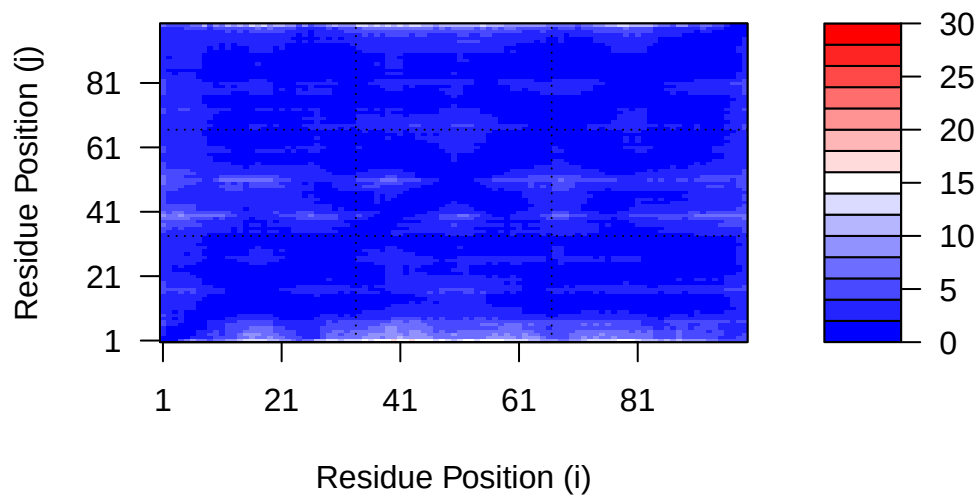
```
plot.dmat(pae1$pae,  
          xlab = "Residue Position (i)",  
          ylab = "Residue Position (j)")
```



```
plot.dmat(pae5$pae,  
          xlab = "Residue Position (i)",  
          ylab = "Residue Position (j)",  
          grid.col = "black",  
          zlim = c(0,30))
```



```
plot.dmat(pae1$pae,  
  xlab = "Residue Position (i)",  
  ylab = "Residue Position (j)",  
  grid.col = "black",  
  zlim = c(0,30))
```



```
aln_file <- list.files(path = results_dir,
  pattern = ".a3m$",
  full.names = TRUE)
```

```
aln_file
```

```
character(0)
```

```
aln_file <- list.files(path = results_dir,
  pattern = ".a3m$",
  recursive = TRUE,
  full.names = TRUE)
```

```
aln_file
```

```
[1] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b.a3m"
[2] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_env/bfd.mgnify30.metaeuk30.smag30.a3m"
[3] "hivpr_dimer_94b5b.result/test_94b5b/test_94b5b_env/uniref.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

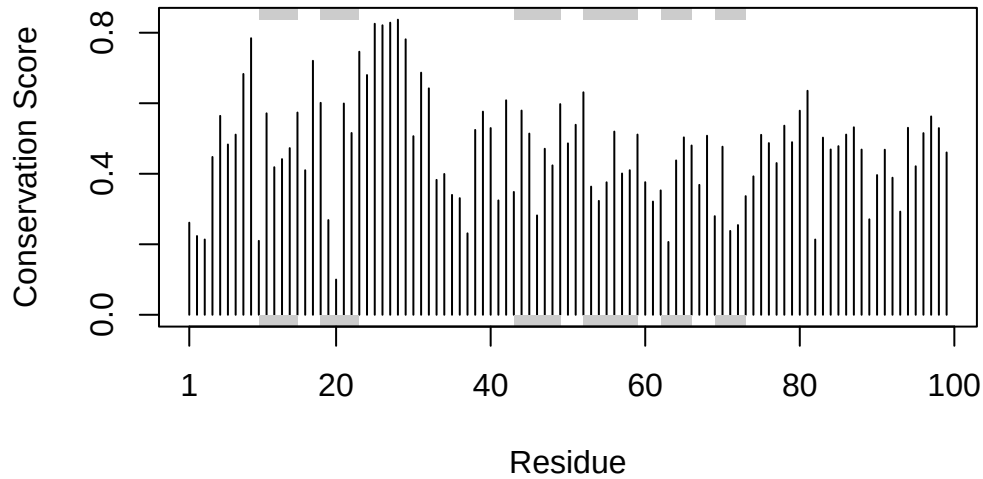
```
[1] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99],  
      sse = trim.pdb(pdb, chain = "A"),  
      ylab = "Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)  
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"  
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-"  
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"  
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"  
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"  
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```



```
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb.files[1])

occ <- vec2resno(c(sim[1:99]), m1.pdb$atom$resno)

write.pdb(m1.pdb, o = occ, file = "m1_conserv.pdb")
```

```
getwd()
```

```
[1] "C:/Users/khach/OneDrive/Desktop/BIMM143/class11"
```

```
file.exists("m1_conserv.pdb")
```

```
[1] TRUE
```

```
write.pdb(m1.pdb, o = occ, file = "./m1_conserv.pdb")
```

```
list.files(pattern = "m1_conserv")
```

```
[1] "m1_conserv.pdb"
```

```
length(occ)
```

```
[1] 757
```

```
nrow(m1.pdb$atom)
```

```
[1] 757
```

```
write.pdb(m1.pdb, o = occ, file = "m1_conserv.pdb")
```

```
list.files()
```

```
[1] "aln.fa" "class11.html"
[3] "class11.qmd" "class11.rmarkdown"
[5] "class11.Rproj" "class11_files"
[7] "corefit_structures" "hivpr_dimer_94b5b.result"
[9] "hivpr_dimer_94b5b.result.zip" "m1_conserv.pdb"
```

```
file.exists("m1_conserv.pdb")
```

```
[1] TRUE
```

The conservation analysis identifies highly conserved residues that may be structurally or functionally important in the HIV protease dimer.